

MAT1100 Homework 1

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Question 3

- (a) *Proof.* First consider $x \in C_{D_8}(r)$. Since D_8 is generated by r and s , there exist $a, b \in \mathbb{Z}$ such that $x = r^a s^b$. Additionally $xr = rx$ which implies:

$$r^a s^b r = r r^a s^b$$

simplifying we get

$$s^b r = r s^b$$

now without loss of generality $b \in \{0, 1\}$ and if $b = 1$ this would imply

$$sr = rs$$

which contradicts the group presentation hence $b = 0$. Therefore

$$C_{D_8}(r) = \langle r \rangle$$

Next consider $x \in C_{D_8}(r)$. $x = r^a s^b$ for $a, b \in \mathbb{Z}$. and since $xs = sx$ we get

$$r^a s^b s = s r^a s^b$$

and using $sr = r^{-1}s$ we find

$$r^a s^b s = r^{-a} s^b s$$

using cancellation

$$r^a = r^{-a} \implies r^{2a} = 1$$

which implies $a = 2 \bmod 4$, i.e. a is even. Therefore

$$C_{D_8}(s) = \langle s, r^2 \rangle$$

□

- (b) *Proof.* First note that $Z(D_8) = C_{D_8}(r) \cap C_{D_8}(s)$. This follows as if an element is in the center it commutes with all of D_8 and in particular r and s ; conversely, since r and s generate D_8 if an element of D_8 commutes with both of them it commutes with the whole group and is hence in the center. Therefore

$$Z(D_8) = \langle e, r^2 \rangle$$

now to determine $Cl_{D_8}(r)$ first note that by orbit-stabilizer (applied to $D_8 \curvearrowright D_8$ by conjugation)

$$|Cl_{D_8}(r)| = \frac{|D_8|}{|C_{D_8}(r)|} = \frac{8}{4} = 2$$

then by the presentation we have $sr = r^{-1}s$ and since $r^{-1} = r^3$ we have

$$sr s^{-1} = r^{-1} = r^3$$

and hence $Cl_{D_8}(r) = \{r, r^3\}$. Similarly

$$|Cl_{D_8}(s)| = 2$$

and by inspection

$$rsr^{-1} = r^2s$$

therefore

$$Cl_{D_8}(s) = \{s, r^2s\}$$

Finally since the center and non-trivial conjugacy classes of D_8 partition it, this implies that the remaining conjugacy classes have size 2. However the remaining elements are not central, and this means their conjugacy classes have size at least 2. Thus there is only one conjugacy class remaining, and it is

$$Cl_{D_8}(rs) = \{rs, r^3s\}$$

□

(c) *Proof.* the class equation for G is:

$$|G| = |Z(G)| + |Cl_{D_8}(r)| + |Cl_{D_8}(s)| + |Cl_{D_8}(rs)|$$

with

$$Z(G) = \{e, r^2\}$$

$$Cl_{D_8}(r) = \{r, r^3\}$$

$$Cl_{D_8}(s) = \{s, r^2s\}$$

$$Cl_{D_8}(rs) = \{rs, r^3s\}$$

□